

# Roof Lifecycle Cost Analysis

Planned maintenance is projected to save the owner of a typical Laurel apartment property **\$274K** over the next 20 years.

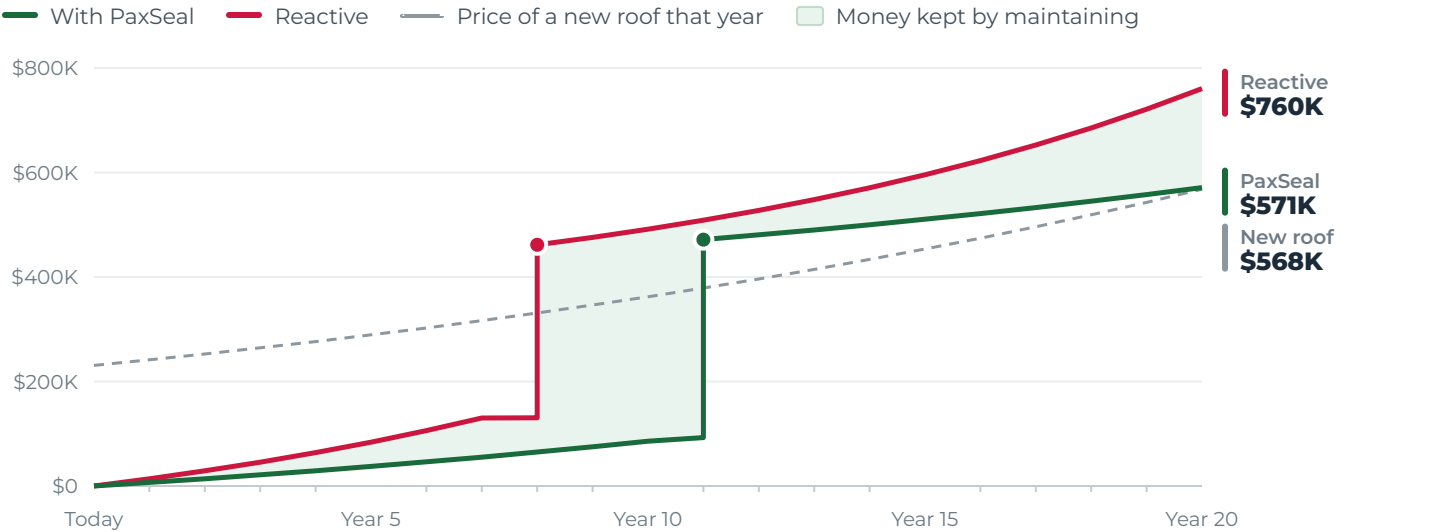
That is 31.6% less than fixing problems only after they leak. Under conservative to upside assumptions, the savings range from \$123K to \$370K. In today's dollars the estimate is worth \$144K.

| MARKET | PROPERTY     | ROOF AREA    | SYSTEM | AGE        | NEW ROOF TODAY |
|--------|--------------|--------------|--------|------------|----------------|
| Laurel | Multi-Family | 30,000 sq ft | TPO    | 8 yrs Good | \$231K         |

|                                                                                                             |                                                                                                              |                                                                                              |                                                                                              |
|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <div>PAXSEAL NET COST</div> <div>\$594K</div> <div>\$571K spent plus roof wear, net of roof life left</div> | <div>REACTIVE NET COST</div> <div>\$868K</div> <div>\$760K spent plus roof wear, net of roof life left</div> | <div>VALUE PROTECTED</div> <div>\$185K</div> <div>\$10K/yr more NOI at a 5.6% cap rate</div> | <div>NEW ROOF PRICE</div> <div>2× in 15 yrs</div> <div>\$231K today, \$568K by year 20</div> |
|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|

## Cumulative roof spending, next 20 years

Nominal dollars. Vertical jumps are full roof replacements.



## Roof life

When each scenario needs a new roof.



Where the savings come from

20-year totals

|                           | PAXSEAL | REACTIVE | SAVINGS |
|---------------------------|---------|----------|---------|
| PaxSeal program fees      | \$126K  | —        | -\$126K |
| Repairs paid by owner     | \$56K   | \$415K   | +\$359K |
| Expected storm damage     | \$10K   | \$14K    | +\$5K   |
| Roof replacements         | \$379K  | \$331K   | -\$48K  |
| Current roof value used   | \$139K  | \$139K   | —       |
| Credit for roof life left | -\$116K | -\$31K   | +\$85K  |
| Net 20-year cost          | \$594K  | \$868K   | +\$274K |

What this means

- 1 **A later replacement.** The next new roof moves from year 8 to year 11, when it costs less in today's dollars.
- 2 **Smaller repairs.** Fixes made on scheduled visits avoid emergency work at about 3.8× the cost in an apartment property, saving \$359K.
- 3 **Fewer unit turns.** A leak into an apartment means displaced residents, mold risk and lost rent. Early repairs keep units occupied.
- 4 **The program pays for itself.** Each \$1 of fees avoids about \$3.17 of other roof costs.

Local risk profile

Prince George's County, MD (24033)

Thunderstorm wind / derecho, snow load

~3.5 major regional roof-damaging events per 20 years (2010 snow, 2012 derecho, 2016 blizzard). Expected storm damage to this roof over 20 years is **\$10K** if maintained and **\$14K** if neglected. A documented maintenance history also helps when an insurer questions a claim.

|                                 |           |       |
|---------------------------------|-----------|-------|
| Hail                            | Moderate  | ■■■■■ |
| Strong wind                     | High      | ■■■■■ |
| Tornado                         | High      | ■■■■■ |
| Hurricane                       | Moderate  | ■■■■■ |
| Winter weather                  | Very High | ■■■■■ |
| Ice storm                       | Moderate  | ■■■■■ |
| FEMA National Risk Index rating |           |       |

Key assumptions

The basis for each figure in this analysis.

| ASSUMPTION           | VALUE             | BASIS                                                                                                 |
|----------------------|-------------------|-------------------------------------------------------------------------------------------------------|
| Roof life            | 20 / 15 yrs       | Maintained / reactive. Fannie Mae service-life tables; +5 yrs per NRC Canada and warranty terms       |
| Roof cost growth     | 4.6%/yr           | BLS Producer Price Index, nonresidential roofing contractors, 2007–2025                               |
| Replacement cost     | \$7.70/sq ft      | Laurel contractor price ranges                                                                        |
| Reactive repair cost | 3× planned × 1.25 | Industry estimates; no primary study found. ×1.25 for leak damage in an apartment property            |
| Storm damage         | 0.13%/yr          | Of roof cost, before insurance. FEMA National Risk Index expected building loss, roof-related hazards |
| PaxSeal fee          | \$4,700/yr        | PAX program estimate for this roof size. Your proposal will state the actual fee                      |
| Cap rate             | 5.6%              | Laurel multi-family market reports                                                                    |

**Estimates, not a quote.** Projections from Laurel market research and industry data, in nominal dollars. The PaxSeal fee is an estimate; your proposal states the actual price. Net cost includes wear on the current roof, less roof life left at year 20. Storm damage is before insurance. Property value is the yearly gain in net operating income divided by the cap rate, for income property where the owner pays for the roof.